

```

vendor-ab-prot8 => PROTOCOL [ 0x00AB0008 ]
vendor-ab-prot8 => PROTOCOL [ 0x00AB:0x0008 ]
vendor-ab-prot8 => PROTOCOL [ MATTER-VENDOR-AB:8 ]

```

B.5.4. length

length Qualifier Syntax

```

type [ length count ]           // exactly count
type [ length min..max ]       // between min and max (inclusive)
type [ length min.. ]          // at least min

```

The **length** qualifier MAY be used to constrain the number of elements in a collection type, such as an **ARRAY** or **LIST**, or the number of bytes in a **STRING** or **OCTET STRING** type.

B.5.5. nullable

nullable Qualifier Syntax

```

type [ nullable ]

```

The **nullable** qualifier is used with **ARRAY**, **LIST**, **STRUCTURE**, **STRING**, **OCTET STRING**, **BOOLEAN**, **SIGNED INTEGER**, **UNSIGNED INTEGER**, **FLOAT32**, **FLOAT64** types. The **nullable** qualifier declares that a TLV **Null** MAY be substituted for a value of the specified type at a particular point in an encoding. For example, in the following **sensor-sample** structure, a null value MAY be encoded for the **value** field (e.g. in the case the sensor was off-line at the sample time):

nullable Qualifier Example

```

sensor-sample => STRUCTURE
{
    timestamp [1] : UNSIGNED INTEGER,
    value [2] : FLOAT64 [ nullable ],
}

```

Applying a **nullable** qualifier to a type is exactly the same as defining a **CHOICE OF** type with **alternates** for the primary and **NULL**. For example, the sensor sample structure could also be defined as follows:

nullable Qualifier Example

```

sensor-sample => STRUCTURE
{
    timestamp [1] : UNSIGNED INTEGER,
    value [2] : CHOICE OF
    {

```